

Matched six- and nine-species comparison

Common chemistry, common feed, and two explicitly separated liquid-thermo lanes

2026-09-01

Table of contents

1	Question and clean sequence	1
2	Coefficient provenance	2
2.1	Ideal-liquid lane	2
2.2	ePC-SAFT liquid lane	3
3	Experiment contract	3
3.1	Ideal canonical-state lane	3
3.2	ePC-SAFT fixed- T, P lane	3
4	Retained results	4
4.1	Ideal lane	4
4.2	ePC-SAFT liquid lane	4
5	Corrected apparent projection and CO₂ pressure	5
6	Parameter reduction and explicit-species follow-up	6
7	All-parameter sensitivity and screening-prior UQ	6
7.1	Parameter universe and sampling contract	6
7.2	Global sensitivity results	7
7.3	Propagated screening-prior intervals	9
8	Interpretation	10
9	Limitations and claim boundary	10
10	Reproduction and retained-file map	11

Working-notebook boundary

This is a working diagnostic notebook. It reads retained CSV tables and rendered figures; it does not rerun the equilibrium calculation. The ideal lane is verified against its solver residual contract. The ePC-SAFT lane is a verified disposable SciPy/EOS calculation using a provisional Engine packet. The all-parameter UQ below is a screening-prior survey, not a measurement- backed posterior. Neither lane is an adopted MEA parameter set or a manuscript result.

1 Question and clean sequence

The question is whether a six-species reduced reaction set gives the same common-species state as the matched nine-species set when the feed, reaction source, temperature, pressure, and reporting basis are held fixed.

The sequence is:

1. Read the source-controlled nine-species reaction correlations.
2. Derive the six-species reaction correlations by exact stoichiometric projection; do not insert a separately fitted six-species parameter set.
3. Solve the nine- and six-species ideal-liquid systems at every canonical VLE and ChEq temperature/loading coordinate.
4. Use the installed ePC-SAFT Engine only for EOS evaluations and source-state transfer. Solve the reaction and balance equations with a disposable SciPy least-squares bridge because the Engine reactive declaration gate does not admit this reduced comparison directly.
5. Compare common-species amounts and report the inventory of carbonate, hydronium, and hydroxide omitted by the six-species set.

The reporting basis is amount per mol initial MEA. Loading is mol CO₂ per mol initial MEA, and the feed is 30 wt% MEA in water.

2 Coefficient provenance

2.1 Ideal-liquid lane

The nine-species ideal coefficients come from the local, source-controlled [reaction constant manifest](#). They are the Edwards/Baygi water-carbonate family, the Tong/Baygi carbamate hydrolysis entry, and the Bates/Pinching/Baygi protonated-amine dissociation entry. The manifest, rather than a fitting-routine output, is the authority for this lane.

The manifest correlation is

$$\ln K(T) = A + B/T + C \ln(T) + DT.$$

The retained nine-species values are:

reaction	A	B (K)	C	D (K ⁻¹)	source key
R1	132.899	-13445.90	-22.4773	0	Edwards1978 via Baygi2015
R2	231.465	-12092.10	-36.7816	0	Edwards1978 via Baygi2015
R3	216.049	-12431.70	-35.4819	0	Edwards1978 via Baygi2015
R4	-1.8652	-1545.30	0	0	Tong2012 via Baygi2015
R5	2.1211	-8189.38	0	-0.007484	Bates1951 via Baygi2015

The six-species reactions are exact combinations of those same five reactions:

$$\begin{bmatrix} R_{6,\text{carb}} \\ R_{6,\text{bicarb}} \end{bmatrix} = \begin{bmatrix} 0 & 1 & 0 & -1 & -1 \\ 0 & 1 & 0 & 0 & -1 \end{bmatrix} \begin{bmatrix} R_1 \\ R_2 \\ R_3 \\ R_4 \\ R_5 \end{bmatrix}.$$

Therefore the derived six-species coefficient rows are:

reaction	source combination	A	B (K)	C	D (K ⁻¹)
carbamate formation	R2 - R4 - R5	231.2091	-2357.42	-36.7816	0.007484
bicarbonate formation	R2 - R5	229.3439	-3902.72	-36.7816	0.007484

These rows are recorded in [reaction_coefficients.csv](#) with `source_status = inference`. The six-species rows are not the earlier sibling fitting-routine coefficients.

2.2 ePC-SAFT liquid lane

For the ePC-SAFT lane, the reaction constants come from the Engine-side MEA source contract used by the candidate input. R1-R3 use the same source forms as above. R4 uses

$$\ln K_{R4} = 2.151 - 1545.3/T,$$

and R5 uses the Bates source form

$$\ln K_{R5} = -\ln(10) (2677.91/T + 0.3869 + 0.0004277T).$$

The ePC-SAFT six-species reaction constants are the same two exact rows of the projection matrix applied to these Engine source constants. This keeps the reaction source separate from the ideal manifest; the ideal R4/R5 rows are not mixed into the ePC-SAFT calculation.

The liquid EOS parameters are not reaction coefficients. They come from the immutable Engine candidate packet recorded in [epcsaft_scipy_packet_manifest.csv](#), which identifies the packet SHA-256 as `12e380a27eb615e25bae580f8c1ec852020879917b08fa1`. The packet contains the common species pure-component, association, Born, dielectric, molecular-weight, and interaction inputs used by both the nine- and six-species EOS evaluations. Its scientific status remains `diagnostic_candidate`; it is not a promoted predictive packet.

3 Experiment contract

3.1 Ideal canonical-state lane

The ideal lane uses every unique coordinate retained in the canonical VLE/ChEq tables: 232 coordinates and 464 model-state rows. For each coordinate the solve enforces the reaction equations, carbon loading, water-to-amine ratio, electroneutrality, and normalization. Activities are dimensionless ideal true-species activities, $a_i = x_i$.

The nine-species order is:

CO2, MEA, H2O, MEAH+, MEACOO-, HC03-, CO3^2-, H3O+, OH-

The six-species order is:

CO2, MEA, H2O, MEAH+, MEACOO-, HC03-

The internal acceptance limit is 10^{-8} for the maximum reaction or balance residual. The retained maximum is 1.989520×10^{-13} and every ideal state converged. Ideal results are in [solve_diagnostics.csv](#), [state_species_results.csv](#), and [common_species_comparison.csv](#).

3.2 ePC-SAFT fixed- T, P lane

The disposable bridge evaluates the Engine EOS at 100 kPa for 293.15, 313.15, and 323.15 K. It solves positive formula-unit extents with SciPy while enforcing the C/N/water feed ratios, exact charge balance, and the transferred reaction affinities. The Engine supplies EOS fugacity coefficients, density, and pressure evaluation; SciPy supplies the temporary equilibrium solve.

For the source-to-EOS standard-state transfer, the bridge uses the Engine neutral-basis representation. In compact form:

$$\frac{\mu_i}{RT} = \ln x_i + \ln \phi_i + \ln \left(\frac{P}{\rho_{ref} RT} \right),$$

and each transferred reaction constant is evaluated as

$$\ln K_{EOS} = \ln K_{source} + \alpha \cdot (c + s) + \Delta \nu \ln \left(\frac{P}{\rho_{ref} RT} \right).$$

The Engine reactive declaration gate is therefore bypassed only for the temporary equilibrium layer; it is not treated as a failed scientific experiment. The retained bridge diagnostics show source transfer passed, charge/balance/reaction residuals below the stated limit, and EOS pressure reproduction within 8.67×10^{-7} Pa. This is a fixed- T, P liquid composition calculation, not a VLE pressure calculation.

4 Retained results

4.1 Ideal lane

quantity	retained result
unique canonical coordinates	232
ideal states attempted	464
ideal states converged	464
maximum raw solve residual	1.989520×10^{-13}
maximum common-species absolute delta	6.510011×10^{-3} mol/mol initial MEA
maximum omitted nine-species inventory	6.510012×10^{-3} mol/mol initial MEA

The maximum ideal omitted inventory occurs at a 20 °C Martin ChEq coordinate with loading 0.31 and is dominated by carbonate. The full coordinate-level inventory is in [omitted_species_inventory.csv](#).

4.2 ePC-SAFT liquid lane

temperature (K)	carbonate amount	hydronium amount	hydroxide amount	total omitted amount
293.15	2.992502×10^{-4}	1.81×10^{-11}	2.408×10^{-9}	2.992526×10^{-4}
313.15	1.555471×10^{-4}	6.01×10^{-11}	2.851×10^{-9}	1.555500×10^{-4}
323.15	1.104485×10^{-4}	1.03×10^{-10}	2.998×10^{-9}	1.104516×10^{-4}

The largest common-species deltas in the retained ePC-SAFT lane are small on this reporting basis: H₂O and MEACOO⁻ reach about -2.86×10^{-4} mol/mol initial MEA, while MEAH⁺ reaches $+2.99 \times 10^{-4}$. The complete rows are in [epcsaft_scipy_comparison.csv](#).

Matched species-set effect on common species

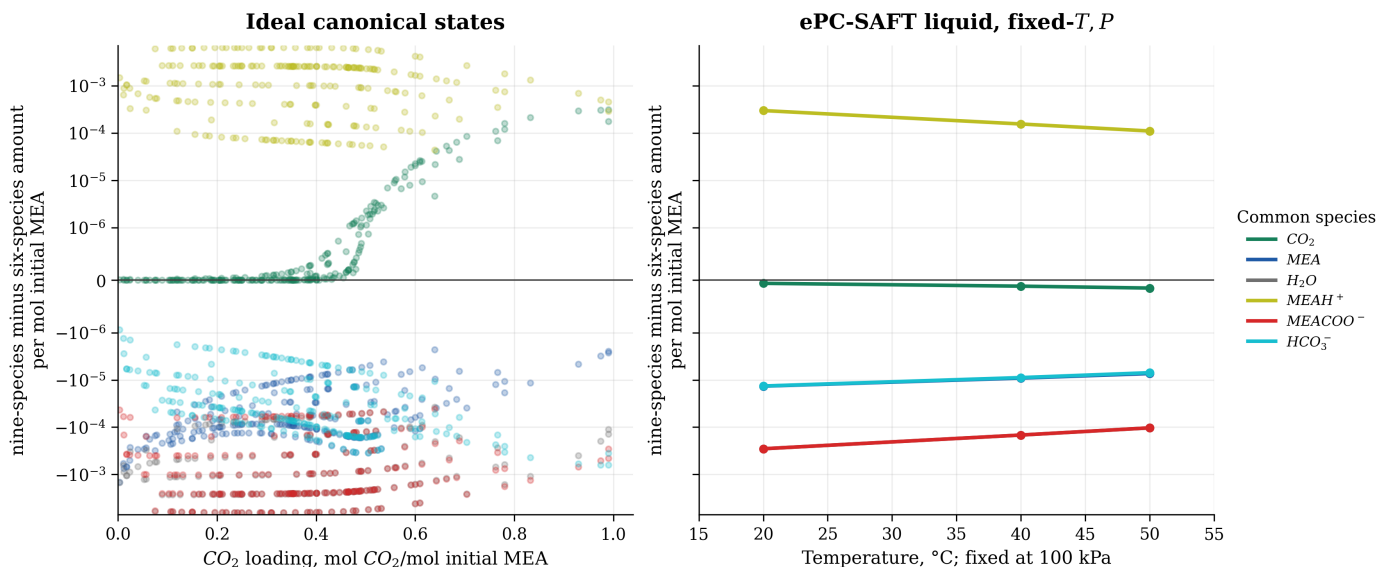


Figure 1: Common-species amount difference between matched nine- and six-species states.

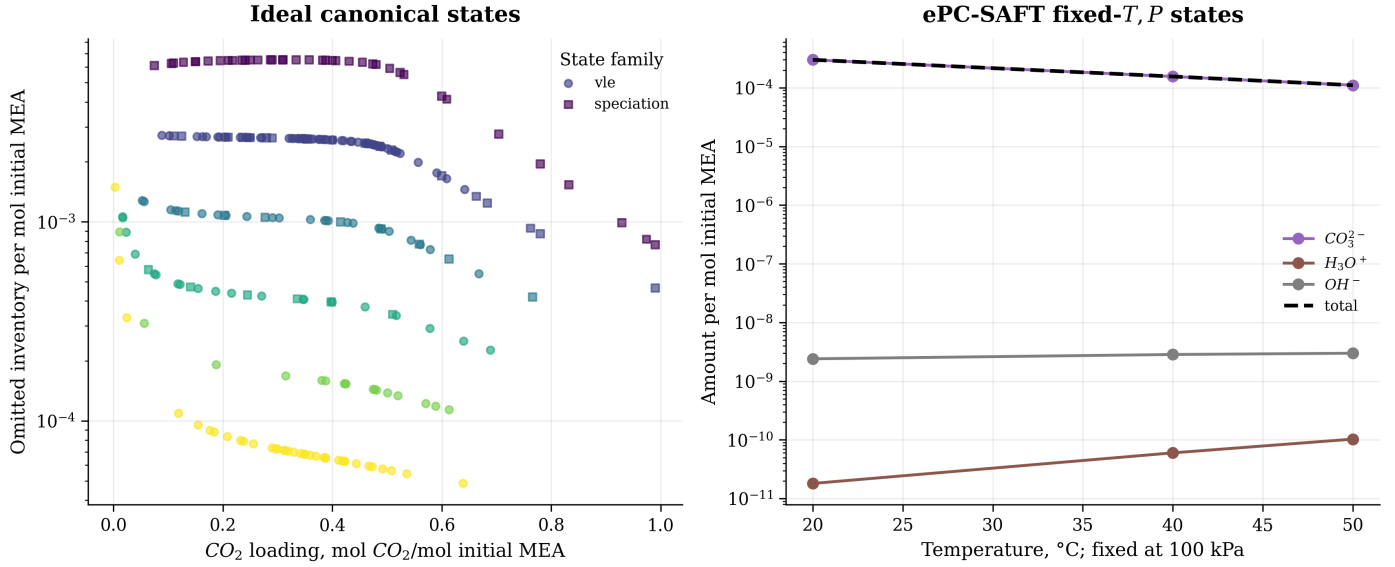


Figure 2: Inventory explicitly omitted by the six-species set.

5 Corrected apparent projection and CO₂ pressure

The corrected neutral projection is formed from conserved totals, not from the first three true-species entries:

$$n_{\text{CO}_2, \text{app}} = n_{\text{CO}_2} + n_{\text{MEACOO}^-} + n_{\text{HCO}_3^-} + n_{\text{CO}_3^{2-}}, \quad n_{\text{MEA, app}} = n_{\text{MEA}} + n_{\text{MEA}^+} + n_{\text{MEACOO}^-},$$

$$n_{\text{H}_2\text{O, app}} = n_{\text{H}_2\text{O}} + n_{\text{HCO}_3^-} + n_{\text{CO}_3^{2-}} + n_{\text{H}_3\text{O}^+} + n_{\text{OH}^-}.$$

The full 161-row CO₂-pressure table was evaluated with this projection and the retained neutral `flashTQ` EOS. The projection equality is numerical to machine precision, so one neutral pressure evaluation is shared by both model labels.

quantity	six species	nine species
pressure rows attempted	161	161
pressure rows converged	1	1
coverage	0.62%	0.62%
median absolute log ₁₀ error on	6.003934	6.003934
converged rows		
maximum	—	1.321165×10^{-14}
projected-composition		
difference		

The single converged row predicted 2038.38 kPa for an observed 0.00202 kPa point. The other 160 rows returned `SolutionError: A solution was not found for flashTQ`. This is a useful negative diagnostic, not a pressure-fit result: the neutral EOS is being asked to treat chemically bound total carbon as free molecular CO₂. A fair CO₂-pressure test therefore needs a reactive bubble calculation with species-specific fugacity, not this neutral apparent flash.

The result also answers the six-versus-nine identifiability question for this observable. At fixed feed and loading, the carbon, amine, and water totals are fixed before the neutral EOS call. Reaction coefficients and internal ionic parameters can change true-species allocation, but their corrected apparent pressure sensitivity is structurally zero. The retained screen is in [corrected_apparent_parameter_sensitivity_screen.csv](#).

6 Parameter reduction and explicit-species follow-up

The ten reactive/ionic identities in the screen are all marked `structural_zero` for corrected neutral pressure. The pressure-only active set should therefore exclude them: keep one-segment counts fixed, keep Born diameters fixed, keep source-bound reaction correlations fixed, and do not fit the provisional MEAH^+ or MEACOO^- size/energy terms against pressure alone. The two ion-size terms remain candidates only for a future explicit-species ePC-SAFT fit against density and speciation with regularization.

This corrected-pressure result is a structural screen, separate from the screening-prior UQ reported below. The retained explicit-species sensitivity studies show that ion-coefficient and Born-form choices can change speciation and can affect convergence, but those studies use older Engine wheel hashes (`e142888...` and `cfe8e2...`) and are not a current packet UQ. They support keeping the full nine-species chemistry for reactive speciation work; they do not justify deleting carbonate or promoting any ion parameter.

7 All-parameter sensitivity and screening-prior UQ

This survey tests the complete continuous thermodynamic parameter inventory, including records that were fixed in the candidate packet. It uses the valid candidate-3 parameter document and the explicit Engine wheel recorded in [uq_run_receipt.json](#). The wheel hash is `sha256:05cca73463b9b44dcb3a77dfd239b767ec63aafce7d18b85b2a66b80773847a5`; the locked production hash is different, so these are diagnostic results.

7.1 Parameter universe and sampling contract

The retained [parameter inventory](#) contains 143 identities:

disposition	count	treatment
included continuous identities/groups	113	varied in the LHC and ranked by Spearman, rank-SRC, and ridge-PRCC
same-sign ionic k_{ij}	7	excluded structurally because the installed ionic topology omits them
derived MEA–water association records	4	excluded because the Engine derives them by a combining rule
chemical identity records	18	fixed by chemistry; not valid continuous UQ coordinates
discrete ionic topology choice	1	retained as scenario metadata, not forced into a continuous LHC

The 113 included groups contain 29 included binary k_{ij} terms, one reciprocal-temperature slope, five reaction-correlation coefficients, and the pure-component association, segment, dispersion, dielectric, Born, Debye–Hückel, solvation, and suppression records. Packet-fixed values were still varied for screening. The ranges are declared screening priors: mostly family-relative intervals (5% for segment counts, 8–20% for most pure and ionic terms, 25% of the absolute value or 0.01 for included ($k_{\{ij\}}$), and unit-aware 20% reaction-coefficient intervals). They are not experimental standard deviations and are not a posterior covariance.

The design is a scrambled, random-coordinate-optimized Latin hypercube with 128 points and seed 20260901, evaluated at the central 313.15 K, 100 kPa, loading-0.29 state. Its largest pairwise input-rank correlation is 0.359. The Engine supplies fugacity coefficients, density, source-reference transfer, and exact fixed-pressure composition derivatives; a temporary SciPy/Newton reaction-extents bridge solves the five equilibrium extents. The equilibrium gate is 10^{-6} in maximum log-affinity residual. Five rows failed the line-search gate and remain in the retained sample table; 123 of 128 rows are valid (96.1% coverage).

7.2 Global sensitivity results

The primary ranking is ridge-PRCC after rank-transforming the LHC inputs and outputs; bootstrap 95% intervals for the unadjusted Spearman coefficient are also retained. The top drivers are:

output	leading parameter groups and ridge-PRCC
$\log_{10} f_{CO_2}$	R5 (a_K) -0.935; water segment-diameter exponent coefficient -0.863; R4 (b_K) +0.605; CO2 dispersion energy -0.541; MEA–MEA H^+ (k_{ij}) -0.539
density	water segment-diameter exponent coefficient -0.956; MEA H^+ dispersion energy -0.637; carbonate Born/Debye–Hückel diameter +0.622; MEA self-association volume +0.544; hydronium Born diameter -0.526
$\log_{10} n_{MEA H^+}$	R5 (a_K) +0.914; hydroxide Born diameter +0.601; MEA dispersion energy +0.544; MEA H^+ –water (k_{ij}) +0.479; CO2–MEA reciprocal slope -0.475
$\log_{10} n_{MEACOO^-}$	R4 (b_K) -0.828; water segment-diameter amplitude +0.574; hydroxide Debye–Hückel diameter -0.569; CO2–water (k_{ij}) -0.566; carbonate–water (k_{ij}) -0.547
$\log_{10} n_{CO_3^{2-}}$	R5 (a_K) +0.968; R4 (b_K) +0.715; water segment-diameter exponent coefficient -0.708; carbamate–hydronium (k_{ij}) +0.560; bicarbonate–CO2 (k_{ij}) -0.522

These ranks answer the binary-interaction question directly: binary terms are not uniformly negligible. CO2–water, carbonate–water, MEA–MEA H^+ , MEA H^+ –water, carbamate–hydronium, and bicarbonate–CO2 interactions enter the leading observable-specific ranks. Their importance is conditional on this provisional packet and declared prior; it does not by itself justify fitting every binary term.

Top all-parameter global sensitivity drivers

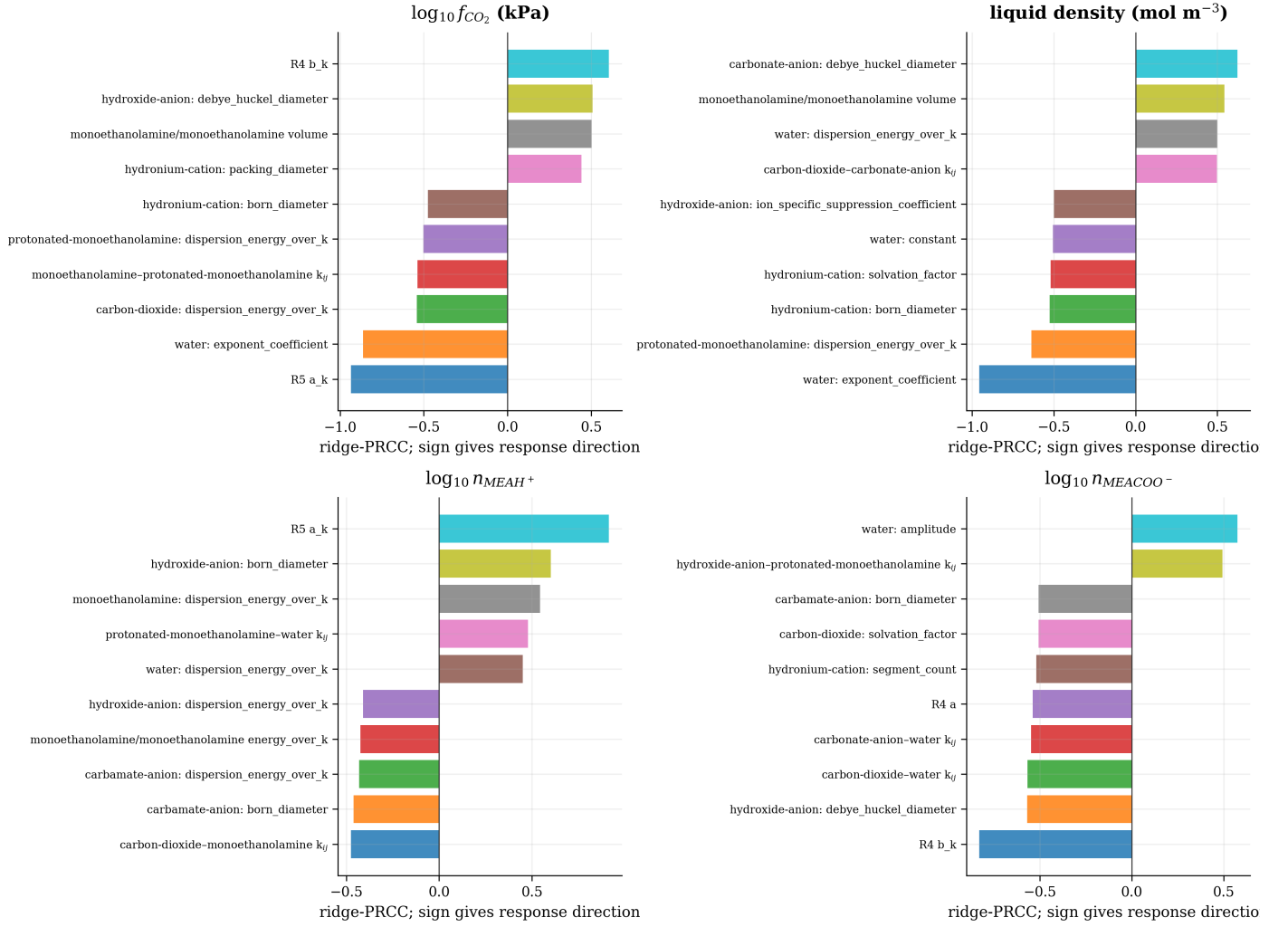


Figure 3: Top all-parameter global sensitivity drivers from the 128-point LHC.

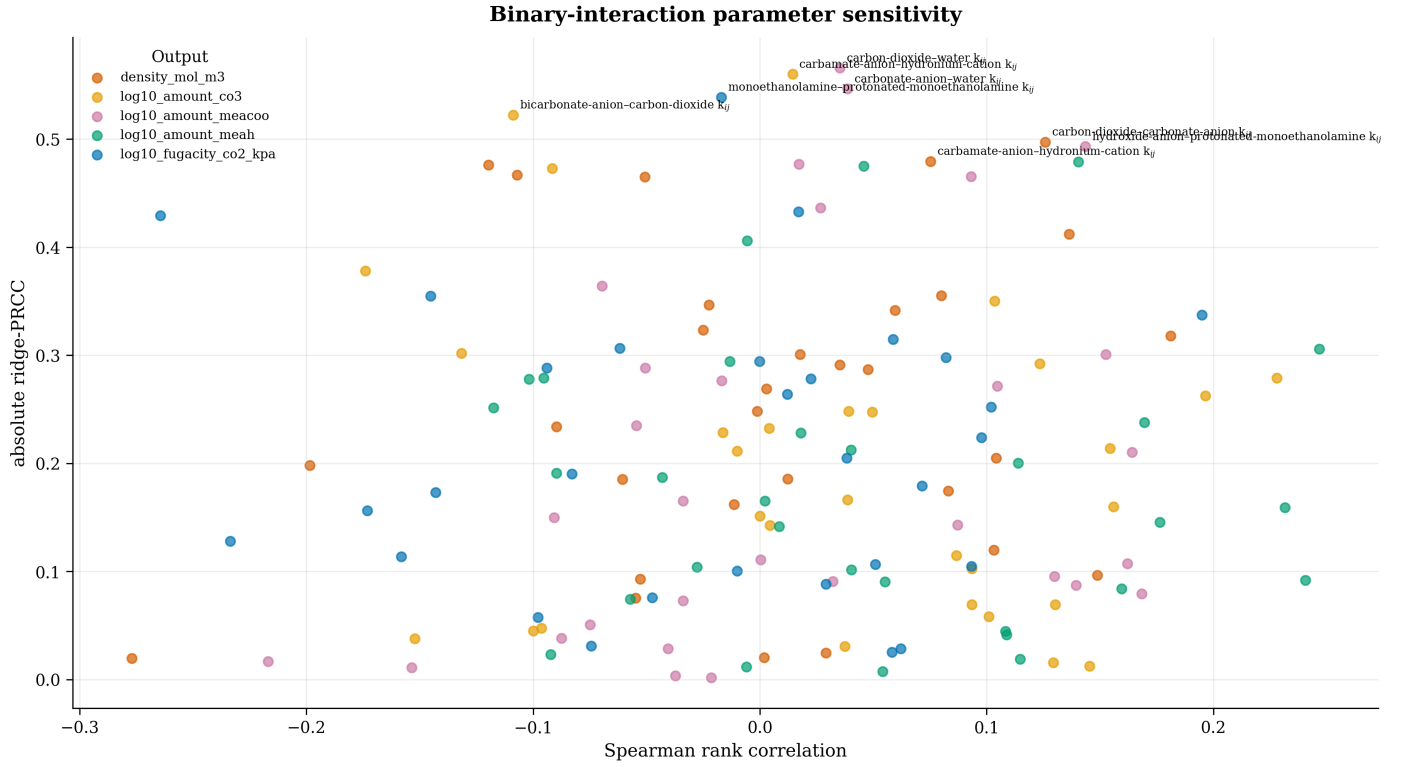


Figure 4: Included binary-interaction groups across all retained outputs.

7.3 Propagated screening-prior intervals

output	baseline	2.5%	median	97.5%	valid/attempted
$\log_{10} f_{CO_2}$ (kPa)	-2.873	-6.254	-3.504	-0.347	123/128
density (mol m^{-3})	49,077	1,049	35,818	68,279	123/128
$\log_{10} n_{MEA H^+}$	-0.538	-0.579	-0.538	-0.530	123/128
$\log_{10} n_{MEA COO^-}$	-0.556	-0.950	-0.556	-0.542	123/128
$\log_{10} n_{CO_3^{2-}}$	-3.818	-6.681	-4.218	-2.272	123/128

The density and CO₂-fugacity ranges are broad because the survey deliberately propagates uncertainty in all EOS, interaction, and reaction coordinates. The reported intervals are screening-prior propagation only—not confidence intervals, credible intervals, or uncertainty on measured data.

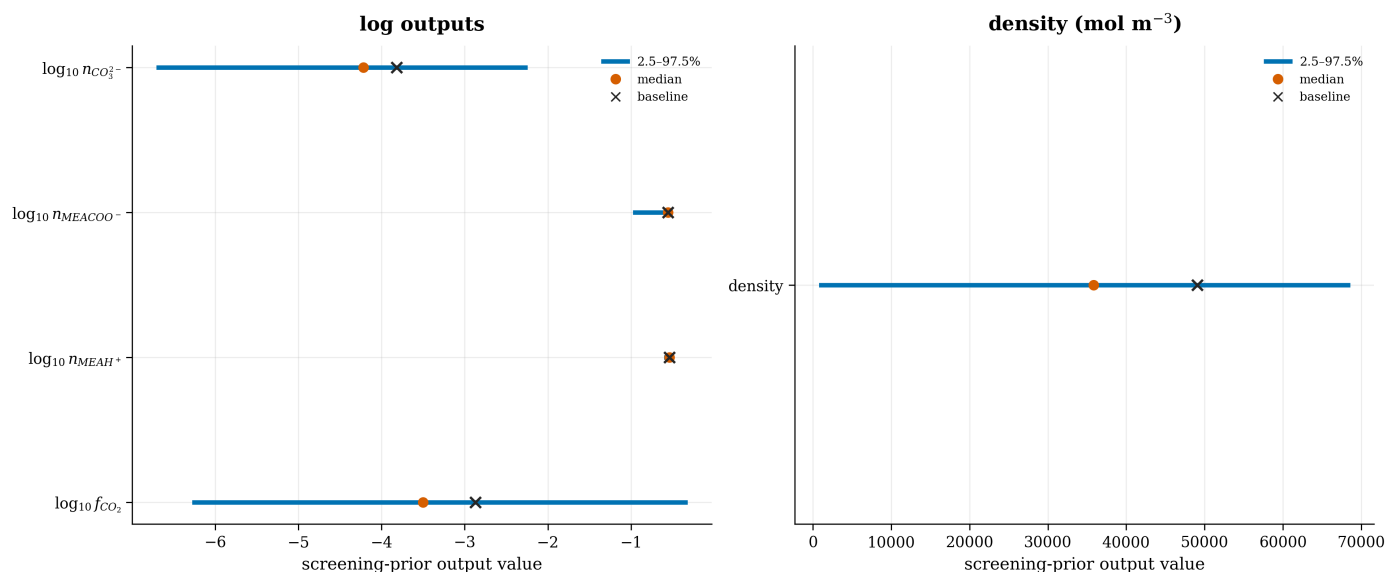


Figure 5: Screening-prior propagated output intervals at the central state.

The optional one-at-a-time endpoint and top-parameter state-grid phases were not run to completion in this execution; their status is retained in [uq_endpoint_status.csv](#). Therefore the global LHC rankings above are the authoritative sensitivity result here, and no missing endpoint/grid rows are interpreted as zero sensitivity.

8 Interpretation

The six-species reduction is not equivalent to simply deleting three columns from a nine-species result. It changes the equilibrium chart and redistributes material among the common species. In the ideal canonical sweep, the largest redistribution is about 6.51×10^{-3} mol per mol initial MEA. In the fixed-100-kPa ePC-SAFT lane, the redistribution is smaller over the three temperatures tested, while carbonate remains the dominant omitted species.

Those statements are retained model comparisons, not causal estimates of a pressure or experimental-error mechanism. The ideal and ePC-SAFT lanes also use different reaction-source contracts and are therefore not pooled into a single score.

9 Limitations and claim boundary

- The ePC-SAFT packet is provisional and diagnostic, not an adopted parameter set.
- The disposable SciPy bridge is a temporary equilibrium method, not a production Engine API or a replacement for Engine declaration validation.
- The ePC-SAFT results are fixed- T , P liquid compositions. They do not claim VLE pressure, bubble-point agreement, global equilibrium, or phase stability.
- No reaction coefficients were fitted during this experiment. The new UQ is screening-prior propagation with rank-based global sensitivity, not a measurement-backed posterior or Hessian-derived uncertainty. The retained corrected-pressure screen remains a separate structural-sensitivity result.
- The six-species ideal lane is not the earlier sibling fitting-routine fit; it is an exact projection of the source-controlled nine-species reaction basis.
- Omitted inventory is reported as a diagnostic. It is not by itself a sensitivity derivative or an attribution of model error.

10 Reproduction and retained-file map

The ideal retained tables can be regenerated with:

```
uv run python analyses/enrtl_six_species_ideal_comparison/scripts/generate_data.py
uv run python analyses/enrtl_six_species_ideal_comparison/scripts/generate_corrected_apparent_pressure.py
uv run python analyses/enrtl_six_species_ideal_comparison/scripts/generate_corrected_apparent_sensitivity.py
```

The claim-led figures can be rendered from retained tables with:

```
uv run python analyses/enrtl_six_species_ideal_comparison/scripts/render_figures.py
```

The all-parameter UQ is checkpointed in four-row LHC parts so the exact 128-point design can be resumed safely. Set `EPCSAFT_ENGINE_WHEEL` to the explicit wheel used for the run and repeat the `lhcb` command over `MEA_UQ_BATCH_START/MEA_UQ_BATCH_STOP` until `[0,128)` is covered:

```
EPCSAFT_ENGINE_WHEEL=/path/to/epcsaft.whl MEA_UQ_PHASE=lhc \\\
  MEA_UQ_SAMPLE_COUNT=128 MEA_UQ_BATCH_START=0 MEA_UQ_BATCH_STOP=4 \\\
  uv run --with /path/to/epcsaft.whl \\\
  python analyses/enrtl_six_species_ideal_comparison/scripts/generate_full_uq.py
EPCSAFT_ENGINE_WHEEL=/path/to/epcsaft.whl MEA_UQ_PHASE=aggregate-lhc \\\
  MEA_UQ_SAMPLE_COUNT=128 uv run --with /path/to/epcsaft.whl \\\
  python analyses/enrtl_six_species_ideal_comparison/scripts/generate_full_uq.py
EPCSAFT_ENGINE_WHEEL=/path/to/epcsaft.whl MEA_UQ_PHASE=finalize \\\
  MEA_UQ_SAMPLE_COUNT=128 MEA_UQ_SKIP_ENDPOINTS=1 \\\
  uv run --with /path/to/epcsaft.whl \\\
  python analyses/enrtl_six_species_ideal_comparison/scripts/generate_full_uq.py
uv run python analyses/enrtl_six_species_ideal_comparison/scripts/render_uq_figures.py
```

The ePC-SAFT bridge was intentionally disposable. Its retained authority is the output packet under `results/epcsaft_scipy_*.csv` and the accompanying [lane status](#), diagnostics, reaction values, packet manifest, and summary. This notebook does not rerun that bridge.

The source and result hashes are retained in [source_manifest.csv](#), [epcsaft_scipy_packet_manifest.csv](#), and [the figure source manifest](#).